

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2025-26 (NEP)

COMPUTER SCIENCE**Paper Code: CSCMJ42 : Data Base Management Systems**

Time: 2:30 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions.***Question 1.**

- (a) Write short notes on any three of the following: [6]
- (i) Data Warehouse
 - (ii) Functionalities and characteristics of DBMS
 - (iii) Database users
 - (iv) Recursive relationship type in ER model
- (b) What do you understand by data models? Write different categories of data models. [4]
- (c) Explain the following: [4]
- (i) Key constraint
 - (ii) Entity integrity constraints
 - (iii) Referential integrity constraint
 - (iv) Disjointness and completeness constraint on specialization/generalization
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Question 2.

- (a) Give examples of the following types of specialization/generalization: [8]
- (i) Disjoint, Total
 - (ii) Disjoint, Partial
 - (iii) Overlapping, Total
 - (iv) Overlapping, Partial
- (b) Explain Theta JOIN, EQUIJOIN and NATURAL JOIN with suitable examples. [6]
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Question 3.

- (a) Explain unary relational algebra operations with suitable examples of each. [9]
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- (b) Write relational algebra operations for the following queries: [5]
- (i) Compute the count (number) of employees and their average salary.
 - (ii) Compute the count of employees and average salary per department.
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Question 4.

- (a) Explain the concept of views and indexes in SQL. How are they created and used? [8]
- (b) Explain different anomalies (update, insert and delete) with their examples. [6]
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Question 5.

- (a) Define Query Optimization and explain its objectives. [4]
- (b) Explain the working of Cost-Based Query Optimization and Heuristic Query Optimization. Describe the factors used in cost estimation. [10]
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Question 6.

- (a) Explain minimal sets of Functional Dependencies. What do you understand by Full functional dependency? Give an example of Full functional dependency. [6]
- (b) Consider the relation $R(A, B, C, D, E, F)$ with functional dependencies $F = \{A \rightarrow BC, C \rightarrow D, B \rightarrow DE, E \rightarrow F\}$. Answer the following: [8]
- (i) Find all candidate keys
 - (ii) Normalize the relation up to BCNF.
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Question 7.

- (a) Explain Lost update and Temporary update problems with their examples in the context of concurrent execution of transactions. Draw state transition diagram to illustrate the states for transaction execution. [7]
- (b) What do you understand by conflict serializable schedule? Write an algorithm to test for conflict serializability. [7]
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